In each discussion, you will have to finish each of the following problems. Use your notes and ask questions if you have any questions!

The link to the book is found on my website, get into the habit of going to the book and doing the problems there! This book is usually what Professor Paulin and other math professors would use to assign problems!

Book Problems:

Section 2.2:

- #5,7,8,12,31-33,44a

Suppose that the position of a particle on the number line at time t is given by the function $x(t) = t^3 + 3t^2$. Find the average velocity of the particle in the time interval $[0, 4]$.

Find the slope of the secant line to $w(x) = 3x^3 + x^2 + 1$ between $x = 0$ and $x = h \neq 0$. Can you express this slope as a polynomial expression in h? What value would it approach as h approaches zero?

Consider the following graph of a function $f(x)$. Estimate for which values of x, the slope of the corresponding tangent line is zero. Estimate the regions on which the slope of the tangent line to f is positive and negative.